

Отчёт к лабораторной работе №6 по предмету «Администрирование в ОС Linux» «Восстановление загрузки операционной системы после сбоя» Вариант 4

Исходное состояние ВМ: загрузка системы не доходит до графического рабочего стола, тк в процессе старта наблюдается циклический отказ lightdm.service

```
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
[ OK ] Started polkit.service - Authorization Manager.
[ OK ] Stopped lightdm.service - Light Display Manager.
Starting lightdm.service - Light Display Manager...
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
[ OK ] Started udisks2.service - Disk Manager.
[ OK ] Stopped lightdm.service - Light Display Manager.
Starting lightdm.service - Light Display Manager...
[ OK ] Finished e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots.
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
[ OK ] Stopped lightdm.service - Light Display Manager.
Starting lightdm.service - Light Display Manager...
[ OK ] Finished logrotate.service - Rotate log files.
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
[ OK ] Stopped lightdm.service - Light Display Manager.
Starting lightdm.service - Light Display Manager...
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
[ OK ] Stopped lightdm.service - Light Display Manager.
[FAILED] Failed to start lightdm.service - Light Display Manager.
See 'systemctl status lightdm.service' for details.
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
```

Диагностика показала, что система загружается в graphical.target, однако сервис lightdm аварийно завершается и уходит в цикл перезапуска. В журнале фиксируется сообщение «Oh no. X Server... It's broken» - это подтверждает проблему в графической подсистеме запуска display manager

```
root@d12:~# systemctl get-default
graphical.target
root@d12:~# systemctl status lightdm --no-pager
* lightdm.service - Light Display Manager
   Loaded: loaded (/lib/systemd/system/lightdm.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Tue 2026-04-07 00:34:40 MSK; 24min ago
     Docs: man:lightdm(1)
   Process: 565 ExecStart=/usr/sbin/lightdm (code=exited, status=234)
   Main PID: 565 (code=exited, status=234)
     CPU: 5ms
file:///d12/lib/systemd/system/lightdm.service (cmd + click)

anp 07 00:34:40 d12 systemd[1]: lightdm.service: Scheduled restart job, restart _at 5.
anp 07 00:34:40 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Start request repeated too quickly.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:40 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Triggering OnFailure= dependencies.
Hint: Some lines were ellipsized, use -l to show in full.
```

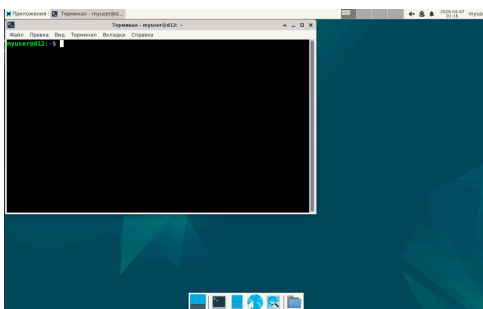
```
root@d12:~# journalctl -u lightdm -b | tail -n 40
anp 07 00:34:38 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
anp 07 00:34:38 d12 lightdm[526]: Oh no. X Server... It's broken
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Main process exited, code=exited, status=234/n/a
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:39 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Scheduled restart job, restart counter is at 1.
anp 07 00:34:39 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
anp 07 00:34:39 d12 lightdm[548]: Oh no. X Server... It's broken
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Main process exited, code=exited, status=234/n/a
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:39 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Scheduled restart job, restart counter is at 2.
anp 07 00:34:39 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
anp 07 00:34:39 d12 lightdm[554]: Oh no. X Server... It's broken
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Main process exited, code=exited, status=234/n/a
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:39 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Scheduled restart job, restart counter is at 3.
anp 07 00:34:39 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
anp 07 00:34:39 d12 lightdm[562]: Oh no. X Server... It's broken
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Main process exited, code=exited, status=234/n/a
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:39 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Scheduled restart job, restart counter is at 4.
anp 07 00:34:39 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:39 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
anp 07 00:34:39 d12 lightdm[565]: Oh no. X Server... It's broken
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Main process exited, code=exited, status=234/n/a
anp 07 00:34:39 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:40 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Scheduled restart job, restart counter is at 5.
anp 07 00:34:40 d12 systemd[1]: Stopped lightdm.service - Light Display Manager.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Start request repeated too quickly.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Failed with result 'exit-code'.
anp 07 00:34:40 d12 systemd[1]: Failed to start lightdm.service - Light Display Manager.
anp 07 00:34:40 d12 systemd[1]: lightdm.service: Triggering OnFailure= dependencies.
```

Выполнено восстановление графической подсистемы: обновлены пакеты Xorg/LightDM, запущен accounts-daemon, удалён пользовательский /etc/lightdm/lightdm.conf

```
root@12:~# apt update
Hit:1 http://deb.debian.org/debian bookworm InRelease (151 kB)
Hit:2 http://security.debian.org/debian-security bookworm-security InRelease (48,9 kB)
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease (52,4 kB)
Hit:4 http://deb.debian.org/debian bookworm/non-free-firmware Sources (7,152 B)
Hit:5 http://deb.debian.org/debian bookworm/main Sources (9,459 kB)
Hit:6 http://deb.debian.org/debian bookworm/main amd64 Packages (8,792 kB)
Hit:7 http://security.debian.org/debian-security bookworm-security/non-free-firmware Sources (796 B)
Hit:8 http://security.debian.org/debian-security bookworm-security/main Sources (385 kB)
Hit:9 http://security.debian.org/debian-security bookworm-security/main amd64 Packages (394 kB)
Hit:10 http://security.debian.org/debian-security bookworm-security/main Translation-en (379 kB)
Hit:11 http://security.debian.org/debian-security bookworm-security/non-free-firmware amd64 Packages (688 B)
Hit:12 http://deb.debian.org/debian bookworm/main Translation-en (6,388 kB)
Hit:13 http://deb.debian.org/debian bookworm/non-free-firmware amd64 Packages (6,368 B)
Hit:14 http://deb.debian.org/debian bookworm/non-free-firmware Translation-en (29,9 kB)
Hit:15 http://deb.debian.org/debian bookworm-updates/main Sources (diff/Index (21,8 kB)
Hit:16 http://deb.debian.org/debian bookworm-updates/main amd64 Packages (diff/Index (21,8 kB)
Hit:17 http://deb.debian.org/debian bookworm-updates/main amd64 Packages (diff/Index (28,7 kB)
Hit:18 http://deb.debian.org/debian bookworm-updates/main Translation-en (diff/Index (2,388 B)
Hit:19 http://deb.debian.org/debian bookworm-updates/main amd64 Packages (6,924 B)
Hit:20 http://deb.debian.org/debian bookworm-updates/main Translation-en (5,448 B)
Получено 25,4 MB за 9с (2,955 kB/с)
Чтение списков пакетов. Готово
Построение дерева зависимостей. Готово
Намет дань обновлению 139 пакетов. Запустите 'apt list --upgradable' для их показа.
Намет обновление в системе. Готово
root@12:~# mv /etc/lightdm/lightdm.conf /etc/lightdm/lightdm.conf.bak 2>/dev/null
root@12:~# systemctl enable --now accounts-daemon
root@12:~# systemctl restart lightdm
root@12:~# systemctl status lightdm --no-pager
● lightdm.service - Light Display Manager
   Loaded: loaded (/lib/systemd/system/lightdm.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-07 01:05:45 MSK; 4s ago
     Main PID: 1348 (lightdm)
       Tasks: 7 (limit: 2294)
        Memory: 85,4M
         CPU: 822ms
    CGroup: /system.slice/lightdm.service
            └─1348 /usr/sbin/lightdm
               └─1354 /usr/lib/xorg/Xorg :8 -seat seat0 -auth /var/run/lightdm/root/:8 -noL...
                  └─1408 lightdm --session-child 13 28

amp 07 01:05:45 d12 systemd[1]: Starting lightdm.service - Light Display Manager...
amp 07 01:05:45 d12 systemd[1]: Started lightdm.service - Light Display Manager...
amp 07 01:05:47 d12 lightdm[1367]: pam_unix(lightdm-greeter:session): session open(id=0)
Hint: Some lines were ellipsized, use -l to show in full.
```

После перезагрузки система успешно загрузилась в графическую среду, выполнен вход, рабочий стол и терминал доступны



Пошаговая инструкция по восстановлению загрузки в графический режим:

1) Проверить целевой режим загрузки и статус display manager:

systemctl get-default

systemctl status lightdm --no-pager

Ожидаем: lightdm.service в состоянии failed

2) Просмотреть журнал ошибок lightdm:

journalctl -u lightdm -b | tail -n 40

Ожидаем: Oh no. X Server... It's broken

3) Обновить индексы пакетов:

apt update

4) Переустановить ключевые пакеты графической подсистемы и дисплейного менеджера:

apt install --reinstall -y lightdm lightdm-gtk-greeter accountsservice xserver-xorg-core

xserver-xorg xserver-xorg-video-all x11-xserver-utils

5) Включить и запустить accounts-daemon:

systemctl enable --now accounts-daemon

systemctl status accounts-daemon --no-pager

6) Убрать локальный конфликтующий конфиг lightdm, если присутствует:

mv /etc/lightdm/lightdm.conf /etc/lightdm/lightdm.conf.bak 2>/dev/null

7) Перезапустить lightdm и проверить состояние:

systemctl restart lightdm

systemctl status lightdm --no-pager

Ожидаем: Active: active (running)

8) Перезагрузить систему:

reboot